Points to be plotted :

x	2	4	6	8	...
y	6	4.5	3	1.5	...

While discussing this with her teacher, the teacher suggested that she can get a unique answer if she solves both the equations together.

By solving equations (1) and (2) we have the cost of an eraser as ₹2 and cost of a pencil as ₹6. It can be visualised in the graph.

The equations we consider together in such settings make a meaningful situation and are known as simultaneous linear equations.

Thus a system of linear equations consists of two or more linear equations with the same variables. Then such equations are called Simultaneous linear equations or System of linear equations or a Pair of linear equations.

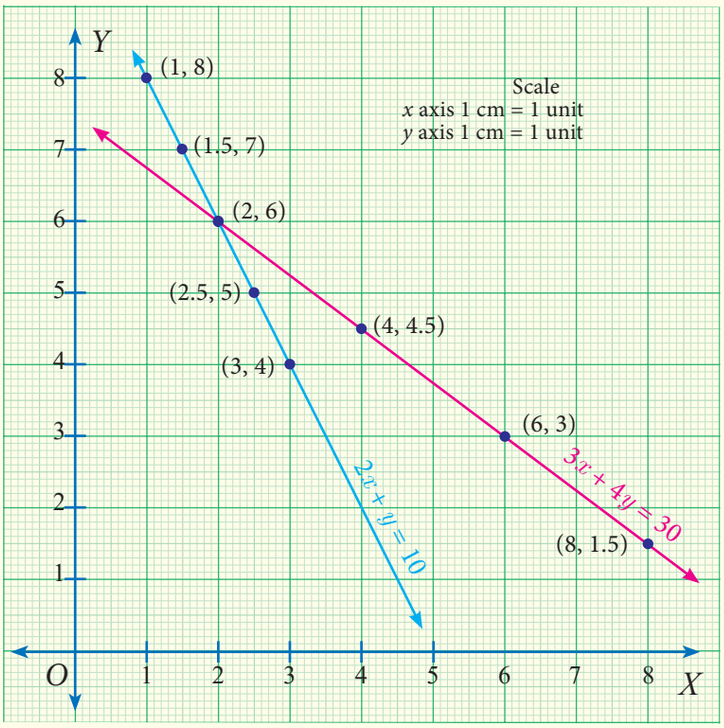


Fig. 3.20

Example 3.43 Check whether (5, −1) is a solution of the simultaneous equations $x - 2y = 7$ and $2x + 3y = 7$.

Solution

Given $x - 2y = 7$ ...(1)

$2x + 3y = 7$ ...(2)

When $x = 5$, $y = -1$ we get

From (1) $x - 2y = 5 - 2(-1) = 5 + 2 = 7$ which is RHS of (1)

From (2) $2x + 3y = 2(5) + 3(-1) = 10 - 3 = 7$ which is RHS of (2)

Thus the values x = 5, y = −1 satisfy both (1) and (2) simultaneously. Therefore (5,−1) is a solution of the given equations.

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